

Health chatbot

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IIS-end sem project



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Project Overview and Objectives



This project aims to build a health-focused chatbot capable of interacting with patients through speech or text, extracting relevant symptoms, and generating a structured summary for clinicians. The chatbot supports multiple languages and can switch dynamically based on user preference, improving accessibility and patient engagement. It also addresses real-world issues like miscommunication, language barriers, and manual data entry in healthcare systems.

Objectives



- Reduce clinician workload by automating data collection during patient intake.
- Improve accuracy and consistency in documenting symptoms.
- Enhance patient experience through natural, guided, and multilingual conversations.
- Enable faster and more efficient triage and diagnosis support using AI.
- Integrate speech-to-text and translation pipelines for real-time, multilingual voice interaction.

System Architecture and Integration



- **Frontend:** Google Colab / Console interaction interface.
- **Backend:** Google Gemini API for symptom analysis and dialogue generation.
- **Data Storage:** MongoDB is used
- **Speech Integration:** Whisper model for speech-to-text conversion.
- **Multilingual Support:** Deep Translator for English, Hindi, Spanish, and German.
- **Integration Flow:**
User Input (speech/text) → Speech-to-Text or Direct Text → Translate if needed → Gemini Prompt → Output Summary

Demo Simulation of Patient Interaction [Prepare a video for max of 2 minutes]



- **Scenario 1:** A patient (use any name) interacts with the chatbot using speech/text to report symptoms. The chatbot processes the input and generates a summary report.

https://drive.google.com/file/d/18R_5r4OE4HgIBNTuL0VwxIV8wdtffE_N1/view?usp=sharing

- **Scenario 2:** The chatbot switches between English and Hindi as per the user's language preference during symptom intake.

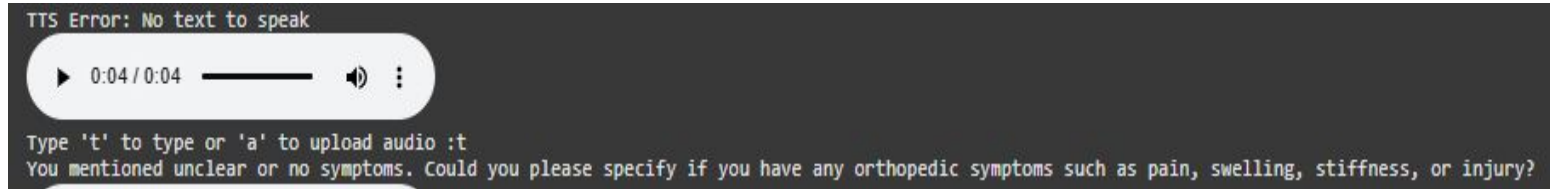
https://drive.google.com/file/d/1zoFK33USZdgfP6OZHM_pWyrTUJ2HHc08/view?usp=sharing

- Make sure to keep the model ready and running on your laptop for any further testing and evaluation.

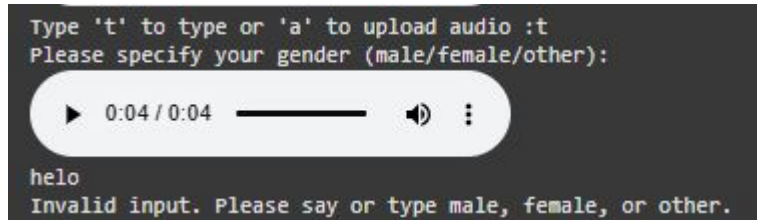
Testing and Evaluation of Features



Miscommunication Recovery: Show how the chatbot handles misunderstood inputs and provides fallback responses.



Real-time Error Handling: Test fallback mechanisms for low-confidence speech recognition or ambiguous input.



- **Challenges & Learnings:**

- Integrating multilingual speech recognition and translation was complex.
- Handling API communication and audio processing required robust error handling.
- Ensuring accurate interpretation of medical symptoms in natural language was challenging due to the variability in patient expression.

- **Future Improvements:**

- Add persistent patient history and medical database integration.
- Enhance condition mapping with medical knowledge graphs and decision trees.

ORTHOPEDIC PATIENT SCREENING REPORT



Patient Information

Name:	Ansh Gahlot
Age:	20
Gender:	Male
Phone Number:	9911557000

Symptoms Summary

Symptom	Location	Duration	Severity	Type	Associated Symptoms
pain in knee	knee	3 days	5	dull	no

Medical History

Condition: no

Follow-up Details:

Overall Summary

Ansh Gahlot, a 20-year-old male (9911557000), presents with a 3-day history of dull, severe (5/10) knee pain. The pain is localized to the knee and is not associated with other symptoms. He reports no significant past medical history. Further evaluation is recommended.